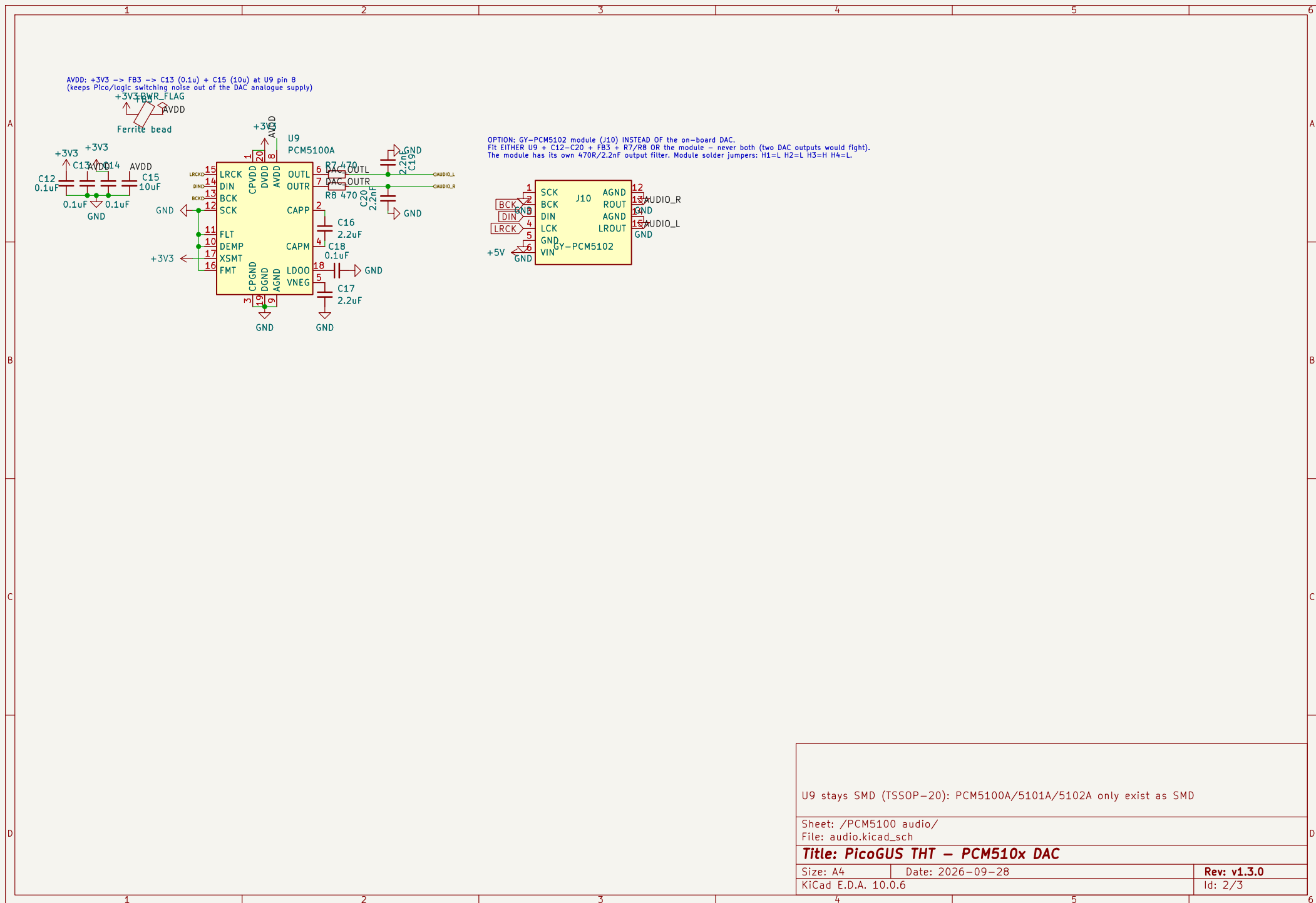




U9 stays SMD (TSSOP-20): PCM5100A/5101A/5102A only exist as SMD

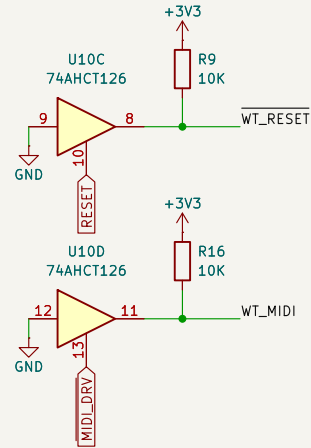
Sheet: /PCM5100 audio/
File: audio.kicad_sch

Title: PicoGUS THT - PCM510x DAC

Size: A4	Date: 2026-09-28	Rev: v1.3.0
KiCad E.D.A. 10.0.6		Id: 2/3

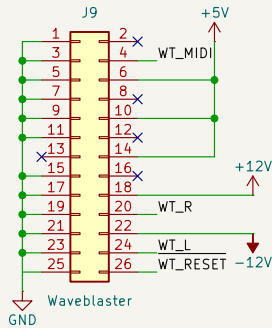
WT CONTROL (U10 = 74AHCT126 @ +5V)

Both outputs are pseudo open-drain (A=GND, OE drives) with pull-ups to +3V3, so the daughterboard never sees 5 V on /RESET or MIDI (some DBs have 3.3 V inputs). /RESET: low while ISA RESET is high. MIDI: same data as the MIDI-out jack (MPU-401 UART).



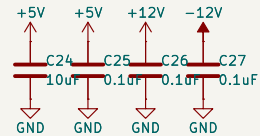
WAVEBLASTER HEADER

2x13 male header, 2.54 mm. Daughterboard (female socket) sits over the card; footprint has the standard standoff holes.

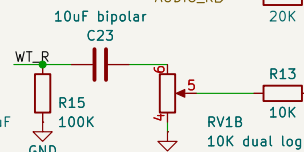
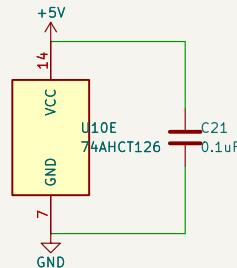


DGND 1-11, AGND 15-25: both to the ground plane

WT SUPPLY DECOUPLING (place at J9)

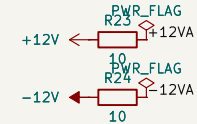
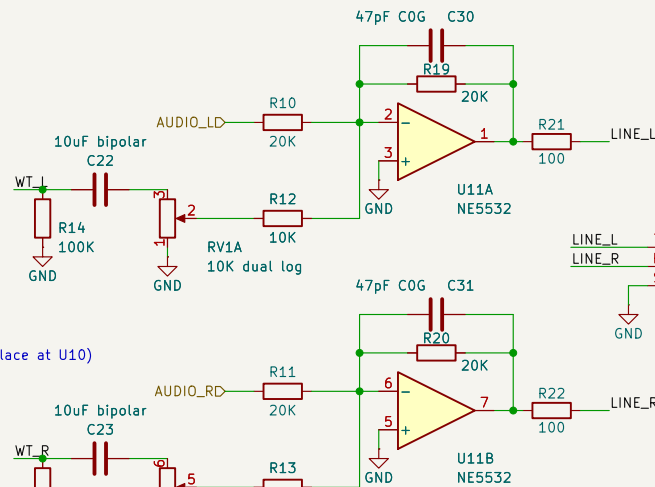


U10 POWER (+5V) + DECOUPLING (place at U10)

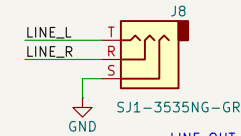
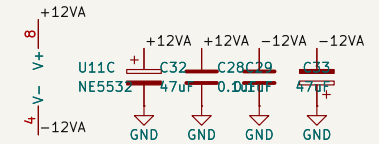


WAVETABLE VOLUME + MIXER

RV1 (dual 10k log) sets the WT level; pins 3/6 = CW end (signal), 1/4 = CCW end (GND). U11 is an inverting summer: PCM 20k in / 20k fb (-unity, -0.2 dB), WT 10k in (x2 = +6 dB at RV1 max; DB outputs are -1-1.2 Vrms vs 2.1 Vrms PCM). The WT knob does not change the PCM level. C22/C23 (bipolar) block DC from the DB. R14/R15 keep them discharged when no DB is fitted. Output: 100R, DC-coupled like v1.2.

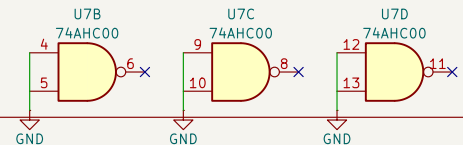


U11 SUPPLY: ISA +/-12 V via R23/R24 (10R) + C32/C33 (47u), C2



LINE OUT (green)

U7 SPARE GATES (74AHC00)



Derived from the hand386 ISA adapter mixer and the PicoGUS 2.0 WT header Waveblaster (wavetable) daughterboard header, analog WT volume and op-amp mixer

Sheet: /Wavetable & line out/

File: wavetable.kicad_sch

Title: PicoGUS THT - Waveblaster header & line-out mixer

Size: A4

Date: 2026-09-28

Rev: v1.3.0

KiCad E.D.A. 10.0.6

Id: 3/3